Marcello Barylli

Education

2025 – Present **PhD at the Robotics, Evolution and Artificial Life Lab** **ITU Copenhagen**

Bio-Inspired Artificial Intelligence
Advisor: Prof. Sebastian Risi
Research: Multi-Agent Systems for Scientific Discovery. Bio-inspired fault tolerance in digital hardware. First-authored work forming the technical basis of an ERC Proof of Concept Grant (*CIRCUIT*, €150k).

2021 – 2024 **MSc, Computational Science, GPA: 8.3/10** (≈ 3.7/4.0 US) **University of Amsterdam**

Thesis: **Multi-Layer Network Models in Colorectal Cancer Subtype Analysis. Grade: 9/10** (≈ 4.0/4.0 US).
Advisor: Dr. Vivek Sheraton Muniraj.
Details: Developed MONIKA, a multi-omic network inference and knockout analysis pipeline.

Key Subjects: Complex systems theory and simulation, machine learning, biosystems data analysis, agent-based models and cellular automata, scientific computing, probability theory.

2017 – 2021 **BSc, Molecular Biology, GPA: 1.7/5** (≈ 3.3/4.0 US) **University of Vienna**

Thesis: **Automated Curation of Genomic Datasets. Grade: 1/5** (≈ 4.0/4.0 US).
Advisor: Prof. Thomas Rattei
Details: Development of software for genomic database filtering and automated grouping.

Key Subjects: Evolutionary theory, developmental biology, systems biology and bioinformatics, quantitative biology, neurobiology, cell culture.

Fellowships

2026 – Present **SPAR Fellow, Biosecurity Track** **SPAR**

Mentor: Noga Aharony (Columbia University). Building a genome language model benchmark that tests frontier models on de Bruijn graph assembly, selecting the correct path at junctions.

Selected Publications

2026 **Barylli, M.**; Béna, G.; Mordvintsev, A.; Nisioti, E.; and Risi, S. **Self-Organising Digital Circuits**. arXiv:2608.02606 [cs.AI].
Basis for *CIRCUIT*, awarded an ERC Proof of Concept Grant (€150k, PI: S. Risi).

Research Experience

2024 **Computational Biologist** **Amsterdam University Medical Center**

Authored scientific journal articles and applied agent-based models in cancer settings.

2023 **MSc Thesis: Multi-Layer Network Models in Colorectal Cancer Subtype Analysis** **University of Amsterdam**

Investigated diffusion-based algorithms for graph neural networks, probabilistic graphical models for network inference, reviewed nonlinear dimensionality reduction techniques for joint embedding of multi-omic data.

Work Experience

2020 – 2021 **Bioinformatician** **CUBE: Computational Systems Biology**

Installation and troubleshooting of software on the Life Science Compute Cluster (LiSC), metagenomic analysis, SQL database implementation and API testing.